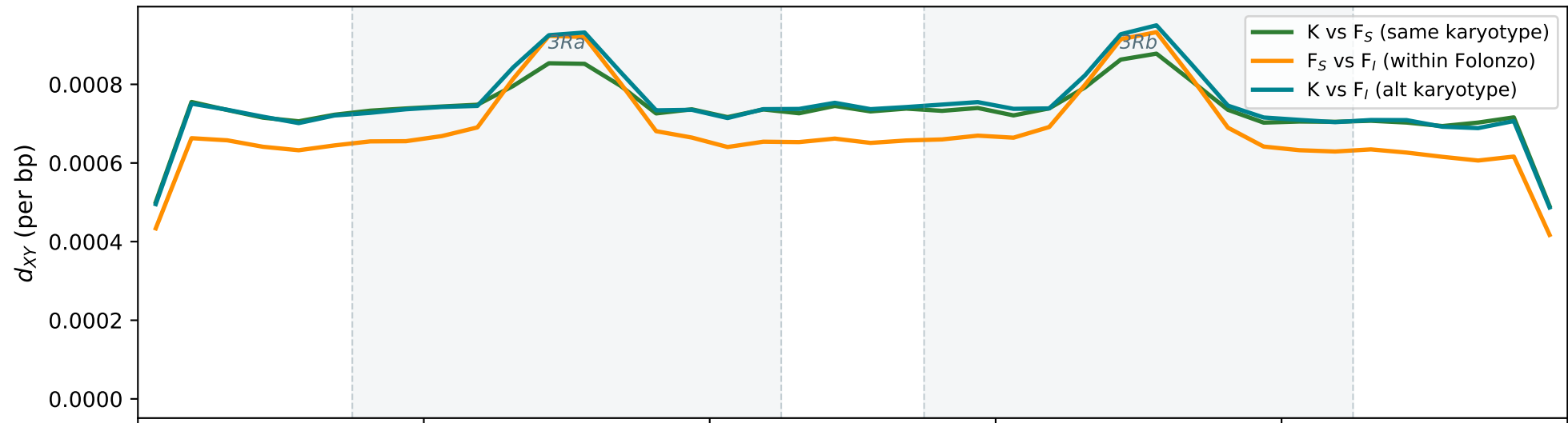


An. funestus Kir/Fol — 3Ra + 3Rb inversion divergence (hull simulator)

A. Absolute divergence d_{XY}



B. Net divergence D_a (same y-axis as A → magnitude visible)

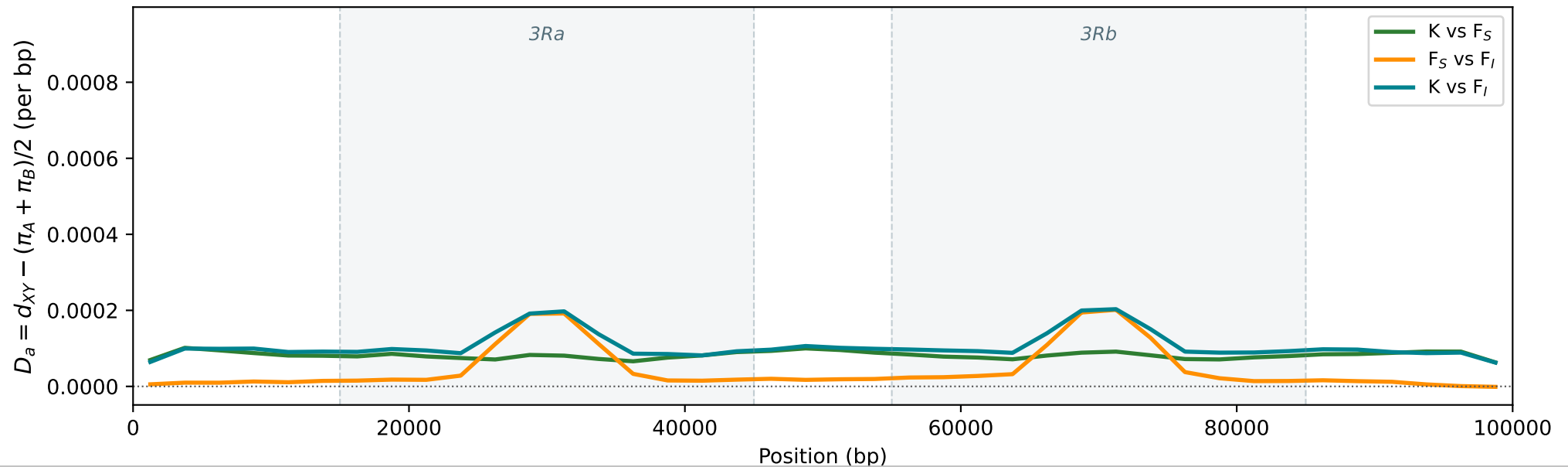


Figure. *An. funestus* Kiribina/Folonzo cross-karyotype divergence for the 3Ra and 3Rb inversions on chromosome arm 3R, simulated with msinv (hull algorithm). (A) Absolute divergence d_{XY} between Kiribina (all-Standard, pop 0) and Folonzo (mixed S/I, pop 1) haplotypes. K vs F_I (alt karyotype) shows elevated d_{XY} inside both inversions due to the recombination barrier, while K vs F_S (same karyotype) shows only the population-split signal. F_S vs F_I (within Folonzo, cross-class) captures the inversion barrier alone. (B) Net divergence D_a on the same y-axis as A — the inversion signal is a small fraction of total d_{XY} , dominated by ancestral polymorphism. Parameters: $Ne_K = Ne_F = 44,000$, $T_{split} = 14,000$ gen, $T_{inv} = 385,000$ gen, $\mu = 3.55e-09$, $r = 4.0e-08$, $\gamma = 0$, 200 replicates.

Command: `pixi run -e all python examples/empirical_kir_fol.py`